

Specifications:

Design and implement a test suite for a Library Alert system with the following specifications:

- UUID resourceID
- string memberEmail
- boolean available

Rules:

- Library resources are identified with UUIDs
- 'checkoutResource' function takes a UUID 'resourceID' and string 'memberEmail' to email a member checking out a book, returning a boolean indicating success or failure
 - 'True' for an available book, then the book status should be updated 'available: False' and an email sent to 'memberEmail'
 - 'False' for an unavailable book
 - 'False' for a null 'resourceID'
- If the 'updateStatus()' or 'sendEmail()' functions fail, a corresponding error should be thrown

→ Analyze Specifications → Identify Causes and Effects

◆ List each distinct **cause** (input condition)

- i. 'resourceID'
- ii. 'memberEmail'
- iii. 'isResourceAvailable'

- iv. 'updateStatus'
- v. 'sendEmail'

◆ List each distinct **effect** (output/action)

- i. updateStatus [True, False]
- ii. sendEmail [True, False]
- iii. return [True, False]
- iv. DatabaseFailureException
- v. EmailFailureException

→ Identify Equivalence Partitions

◆ resourceID Partition(s):

- i. [valid UUIDs]
- ii. NULL: N/A, N/A, F

◆ memberEmail Partition(s):

- i. [well-formed email string]
- ii. NULL: unspecified behavior
- iii. [malformed]: unspecified behavior

◆ isResourceAvailable Partition(s):

- i. [True, False]

◆ updateStatus Partition(s):

- i. [True, False]

◆ sendEmail Partition(s):

i. [True, False]

→ Identify Boundary Values

- ◆ resourceID BVA: N/A
- ◆ memberEmail BVA: N/A
- ◆ isResourceAvailable BVA: N/A
- ◆ updateStatus BVA: N/A
- ◆ sendEmail BVA: N/A

→ Build Decision Table

Cause	Rule #1	Rule #2	Rule #3	Rule #4	Rule #5
rID	[valUUID]	[valUUID]	[valUUID]	[valUUID]	NULL
mEmail	[valEmail]	[valEmail]	[valEmail]	[valEmail]	N/A
isResAv	T	T	T	F	N/A
updateStatus	T	T	F	N/A	N/A
sendEmail	T	F	N/A	N/A	N/A
Effects	T, T, T	T, EmailFailureException, T	DatabaseFailureException, N/A, N/A	N/A, N/A, F	N/A, N/A, F

→ Build TCI Table

TCI	Partitions, Values, Rules
EP01	resourceID: [valid UUIDs]
EP02	memberEmail: [well-formed email string]
EP03	isResourceAvailable: [True, False]
EP04	updateStatus: [True, False]

EP05	sendEmail: [True, False]
EP06	resourceID: NULL
EP07	memberEmail: NULL
EP08	memberEmail: [malformed]
DT01	Rule #1
DT02	Rule #2
DT03	Rule #3
DT04	Rule #4
DT05	Rule #5

→ Build Test Cases

Legend:

- Input #1: resourceID Input #2: memberEmail
- Input #3: isResourceAvailable Input #4: updateStatus
- Input #5: sendEmail
- E1: DatabaseFailureException E2: EmailFailureException

Test Case	TCI(s) Tested	Input #1	Input #2	Input #3	Input #4	Input #5	Expected Output
TC01	EP01, EP02, EP03, EP04, EP05, DT01	[valUUID]	test@test. com	T	T	T	T, T, T
TC02	EP01, EP02, EP03, EP04, EP05, DT02	[valUUID]	test@test. com	T	T	F	T, E2, N/A

TC03	EP01, EP02, EP03, EP04, EP05, DT03	[valUUID]	test@test. com	T	F	N/A	E1, N/A, N/A
TC04	EP01, EP02, EP03, EP04, EP05, DT04	[valUUID]	test@test. com	F	N/A	N/A	N/A, N/A, F
TC05	EP06, DT05	NULL	N/A	N/A	N/A	N/A	N/A, N/A, F
TC06	EP01, EP03, EP04, EP05, EP07	[valUUID]	NULL	T	T	F	T, E2, N/A
TC07	EP01, EP03, EP04, EP05, EP08	[valUUID]	test	T	T	F	T, E2, N/A